

Enabling combined access to electricity and clean cooking with PV-microgrids: new evidences from a high-resolution model of cooking loads

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ABSTRACT

Electric cooking (or e-cooking) based on renewable sources and highly-efficient cooking devices could represent a sustainable and reliable option to achieve the universal access to clean cooking facilities by 2030. Still, the techno-economic feasibility of e-cooking has never been evaluated through (i) a high-resolution assessment of the electric and cooking load profiles, coupled with (ii) a least-cost energy system optimisation for the related electricity supply. Hence, the present study aims at accurately investigating the techno-economic potential of a fully-renewable solar micro-grid, ensuring an integrated access to electricity and clean cooking, based on two representative case studies in Tanzania, namely: (i) a residential case study, and (ii) a community-service case study. Bottom-up stochastic load profiles are generated for the two contexts by expanding the existing LoadProGen model with a novel complementary algorithm for the computation of high-resolution cooking loads. The cost-optimised results prove the cost-competitiveness of e-cooking, especially for the community-service case, where it would require only an additional 17% capital investment due to the pre-existence of other energy-intensive activities. Moreover, the range of Levelised Cost for Cooking a Meal (LCCM) obtained for e-cooking considering the different scenarios lies within $0.16 \div 0.70$ USD/meal, a range that is comparable with that of all other cooking options – including firewood –, and even more cost-competitive than LPG.

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Introduction

According to the recent Energy Access Outlook by the International Energy Agency (IEA), around 2.7 billion people currently lack access to clean cooking facilities while 990 million people lack access to electricity (IEA, 2018). A large share of these people live in rural areas of developing countries, where the lack of a reliable, affordable and clean access to energy contributes to hamper economic growth and social development and entails a reliance on traditional energy sources (such as combustion of biomass, coal and kerosene) that have a severe impact on health and environment (Bailis, Drigo, Ghilardi, & Masera, 2015; Riva, Ahlborg, Hartvigsson, Pachauri, & Colombo, 2018). While significant progresses have been made for electrification, with the rate of people gaining access accelerating up to 100 million per year since 2012, the number of people without access to clean cooking remained globally the same as it was in 2000. In sub-Saharan Africa, the absolute number of people cooking with solid biomass has even increased by 50% since 2000, driven by population growth, thus leading to a total of around 780 million people – 80% of the population – relying on this traditional source (International Energy Agency, 2017).

As a matter of fact, the two sides of access to energy – electrification and clean cooking – are commonly treated as two separate goals, and programmes aimed at the provision of electricity typically do not take into account the inclusion of electric cooking options (International Energy Agency, 2017). Indeed, the efforts of the international community and of national governments in addressing the cooking problem have been focusing so far primarily on the promotion and dissemination of Improved Cooking Stoves (ICSs) (HIVOS, Couture, & Jacobs, 2016; Urme & Gyamfi, 2014), i.e. technologies that are assumed to ensure a cleaner and more efficient combustion (thanks to a more sophisticated design as compared to traditional stoves) while still allowing for the use of traditional fuels (Mehetre, Panwar, Sharma, & Kumar, 2017). However, an increasing number of studies is questioning the effectiveness of ICSs in ensuring the claimed benefits in real-life situations (Sambandam et al., 2015; Wathore, Mortimer, & Grieshop, 2017), and even when partial benefits (such as fuel savings) are achievable, ICSs should be regarded, at best, as an interim solution (HIVOS et al., 2016; International Energy Agency, 2017): in fact, even most advanced models have been proven to be not sufficiently clean to meet the health targets set by the World Health Organisation (WHO) for Household Air Pollution (HAP) (Aung et al., 2016; Pope, Bruce, Dherani, Jagoe, & Rehfuess, 2017; Quansah et al., 2017), and their environmental sustainability in the long term is questionable as well, in consideration of the foreseen population growth in regions already characterised by a massive reliance on traditional biomass (HIVOS

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et al., 2016). For these reasons, some governments are turning their clean cooking efforts on so-called “modern fuels”, especially LPG and natural gas. Nonetheless, those fuels are not renewable either and may conflict with long-term climate change mitigation targets, while their supply chain is typically difficult to extend to rural areas, and less reliable (HIVOS et al., 2016; Nerini, Ray, & Boulkaid, 2017).

Electric cooking (hereafter abbreviated in e-cooking) based on renewable sources and highly-efficient cooking devices would represent an ideal sustainable and reliable option to achieve clean cooking for all, especially in rural and remote areas (Batchelor et al., 2018; Yangka & Diesendorf, 2016). Still, there are very few studies in the literature that accurately consider the possibility of e-cooking, due to the challenges presented by: (i) the large amount of electric power required; (ii) the inherent difficulties in modelling cooking loads; (iii) the cost of electricity; and (iv) socio-cultural barriers (Batchelor et al., 2018; Brown, Leary, Davies, Batchelor, & Scott, 2017; HIVOS et al., 2016; International Energy Agency, 2017; Leach & Oduro, 2015). Nevertheless, especially in a long-term perspective, it is critical to investigate how to overcome such barriers in order to provide a comprehensive and sustainable solution to both the key issues of access to energy (HIVOS et al., 2016; Yangka & Diesendorf, 2016). In fact, there is an untapped potential for renewable e-cooking in relationship with the increasing cost-effectiveness of decentralised systems, especially micro-grids, based on PV and storage systems (Balderrama, Canedo, Fernandez, Lemort, & Quoilin, 2016; Gandini & de Almeida, 2017; HIVOS et al., 2016; International Energy Agency, 2017; Kirchhoff, Kebir, Neumann, Heller, & Strunz, 2016; Mandelli, Barbieri, Mattarolo, & Colombo, 2014). The few studies in the literature assessing the economic potential of e-cooking are based either on different technical configurations (e.g. stand-alone Solar Home Systems, which do not allow for economies of scale) or on data which are not up-to-date, and they are typically not techno-economically optimised – i.e. they only include economic calculations based on the total amount of energy required for cooking and the price of electricity, rather than least-cost optimisations of the system based on the actual load profile (Batchelor et al., 2018; Brown et al., 2017; HIVOS et al., 2016; Nerini et al., 2017). Indeed, an accurate and high-resolution assessment of the load profile for a rural micro energy system is a complex task associated with a high degree of uncertainty, which is possibly exacerbated by the inclusion of cooking loads (Hartvigsson & Ahlgren, 2018; Mandelli, Merlo, & Colombo, 2016); nonetheless, it represents an essential prerequisite to ensure optimal and least-cost sizing of the system (Hartvigsson & Ahlgren, 2018).

The present study thus aims at accurately investigating the techno-economic potential of a fully-renewable solar micro-grid ensuring an integrated access to electricity and clean cooking, based on two representative case studies in Tanzania for which high-resolution electric and cooking load profiles are opportunely modelled: (i) a residential-scale case study based on the surveys carried out in the villages of Kidegeyembe and Nyombo, in the Njombe region; and (ii) a community-service case study based on the Mama Kevina Hope Center, near the city of Same.

Case studies characterisation

Residential case study: Kidegeyembe and Nyombo

Kidegeyembe and Nyombo are two small villages placed around 20–30 km East of Njombe. The villages are characterised by few main roads and productive centres: milling machines, carpentries, garages, shops, guest houses, hair cutters and pubs. Despite the presence of these activities, the main source of income of households is agriculture. The villages were electrified in the end of 2014 by the national utility TANESCO through the extension of the national grid. Based on the assessment carried out by Riva et al. (Riva, Berti, Mandelli, Pendeza, & Colombo, 2017) in 2016 and the information exchanged with the personal of the local NGO CEFA, a community of 33 households was considered for evaluating their electrical load profile with and without the e-cook concept.

Community-service case study: Mama Kevina Hope Center

Mama Kevina Hope Center is a residential structure for children with mild to severe disabilities. The centre has 3 main buildings (Fig. 1): the first one is the dormitory, living room and toilets for the children; in the second one there is the kitchen, the dining hall and the sister's head office; the third one is the dormitory and living room for the sisters. The other smaller buildings are the guest house, the garage, the therapeutic room, the millstone house, the craftsman house and the guardian room. Based on the on-field visit of the Authors in January 2018, the number and type of electric devices used in the buildings, the average daily use, the minimum functioning time and the functioning windows for each appliance were assessed, as well as the information about the cooking habits in the community kitchen.

Methodology

Modelling cooking load profiles

In the absence of measured data, load profiles for off-grid systems in rural areas are typically modelled by means of bottom-up approaches that rely on interview-based information about appliances' ownership and time frames of use (Hartvigsson & Ahlgren, 2018; Mandelli et al., 2016). Mandelli et al. (2016) recently showed how the use of high-resolution stochastic approaches – as the one they proposed and released as LoadProGen – allows to improve the accuracy of such synthetic profiles by embodying in a set of stochastically-generated synthetic profiles the uncertainty and the degree of randomness associated with interview-based data. However, the approach they outlined by Mandelli et al. does not allow to manipulate all the parameters that would be subject to random variations within actual cooking tasks. In fact, cooking cycles (i.e. the sequences of various cooking tasks required to prepare a full meal) (Lombardi, Riva, Bonamini, & Barbieri, 2017) are highly variable as a function of the type of meal cooked and of individual habits and preferences (Ruiz-mercado & Masera, 2015), and a proper stochastic representation of those would require a random variation of both cycles temporal structure and absorbed power. To this regard, we build on the previously mentioned approach by designing a novel model for the stochastic generation of high-resolution cooking loads that interacts with the original LoadProGen model as an outer algorithm, ultimately allowing to modify all the cooking-related parameters (i.e. the range of timings $d_{ij, range}$ that can be associated with each segment of the pre-defined cooking cycle, and the range of power $P_{ij, range}$ level associated to the cooking appliance



Fig. 1. Mama Kevina Hope Centre satellite view.

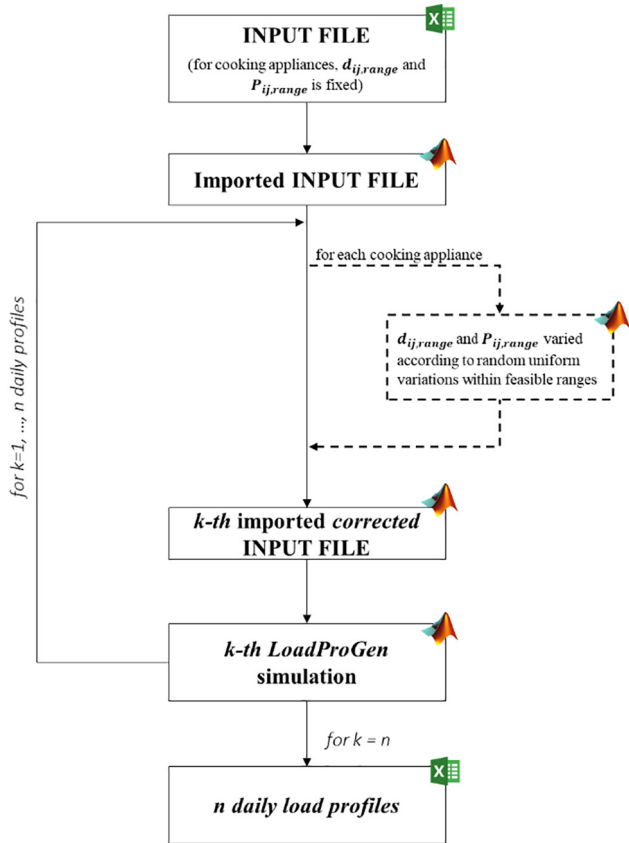


Fig. 2. Conceptual representation of the interaction between the original LoadProGen model and the novel outer script.

in each segment of the pre-defined cooking cycle [W]) according to potential feasible ranges of values (described in the [Cooking devices and cooking habits characterisation](#) sub-section) for each stochastically-generated profile. The interaction between the new outer routine (in bold dotted lines) and the original algorithm (in solid line) is graphically represented in Fig. 2, while the inputs needed by the novel integrated model are reported in Table 1.

Cooking devices and cooking habits characterisation

The load demand associated with each cooking task is a function of both the meal prepared and the cooking device employed. In this study, considering that the object of the investigation is the assessment of the e-cooking potential for a medium- to long-term substitution of traditional fuels, we assume that the cooking devices are efficient

Table 1
Inputs needed by the integrated model for the generation of cooking loads.

j	Specific user class (e.g. “households”)
N_j	Number of user within class j
i	Type of appliance (e.g. light bulb)
n_{ij}	Number of appliances i within class j
P_{ij}	Nominal power [W] of appliance i within class j
h_{ij}	Functioning time [min], or overall time each appliance i within class j is on during a day
$w_{F, ij}$	Functioning windows, or period(s) during the day when each appliance i within class j can be on
d_{ij}	Functioning cycle [min], i.e. minimum continuous functioning time once appliance ij is on (or pre-defined cycle, as an array of minutes associated with different power levels)
Rh_{ij}	% random variation of functioning time for appliance ij
Rw_{ij}	% random variation of functioning windows for appliance ij
$P_{ij, range}$	Range of power level regulations allowed by the appliance ij [W]
$d_{ij, range}$	Range of timings that can be associated with each segment (e.g. cooking task) of a pre-defined cycle

Table 2

Summary of cooking habits in Kidegembye with corresponding timings and induction cooker power ranges (LP: low power; MP: medium power; HP: high power).

	Timing [min]	Power range
Breakfast		
Milk/tea	4–6	HP
Chapati	4–6	MP
Lunch/dinner		
Ugali	20–30	LP
Rice	20–30	LP
Soup	20–40	LP
Vegetables/potatoes/meat/fish	5–10	MP
Water boiling	15–25	HP

induction cookers. In particular, for the residential case, we consider single-plate induction cookers, modelled for simplicity as devices allowing for three power ranges, namely¹: *low-power* (500–1000 W); *medium-power* (1100–1500 W); and *high-power* (1600–2000 W). For the community-service case, instead, we consider a whole 4-plate induction kitchen, with the following power ranges: *low-power* (1000–1400 W); *medium-power* (1500–2000 W); *high-power* (2100–2500 W).

As regards the meals composing typical daily cooking cycles of the region, those are derived from direct field observations in both locations (Kidegembye and Mama Kevina), complemented with interviews with experts from local NGOs and with information from national reports (Tanzania Food and Nutrition Centre and TAHEA Iringa, 2013). In general, the average Tanzanian diet consists of: milk or tea for breakfast, consumed with *chapati* (a type of bread) or fruits; and *ugali* (maize flour cooked in boiling water) or rice as main dishes for lunch and dinner, served with vegetables, beans or (more rarely) fish or meat. The representative cooking tasks and the associated timings and power levels considered for the analysis are reported in Tables 2 and 3 for the two case studies.

Time and power variations relative to single cooking tasks within a cooking cycle are employed by the model to represent the natural variability associated with: a) different cooking habits for each household; b) quantity and typology of food prepared; c) single event (cooking cycle) stochasticity, as further discussed below.

Model of cooking cycles

The developed outer script allows randomising, within the pre-defined ranges reported above, the timing and power level required by each cooking task; subsequently, the aggregation of individual randomised cooking tasks forms a random cooking cycle. The criteria by which cooking tasks are aggregated to form cycles are here exposed for the example of the residential-scale case study, while similar criteria adopted for the Mama Kevina case study are reported in the Supplementary Material.

The cooking cycle associated with breakfast is modelled as a combination of two tasks, in accordance with those identified in.

Table 1, namely: i) heating up milk or water for tea (4–6 min, HP); and ii) preparing Chapati (4–6 min, MP). As a result, breakfast total preparation time will vary between 8 and 12 min (Fig. 3).

Similarly, lunch and dinner have been modelled indiscriminately as a combination of four tasks within those classified as “Lunch/Dinner” tasks in Table 2. Firstly, water or oil are boiled for vegetables/meat/fish preparation (5–10 min, HP) before the latter are cooked for 5–10 min at MP (Tanzania Food and Nutrition Centre and TAHEA Iringa, 2013). Secondly, the main dish preparation is carried out starting with boiling water in a large pot (10–15 min, HP)² and continuing with rice, ugali or

¹ The power ranges and the efficiencies of the modelled devices are estimated based on the datasheets of some commercial product, such as Duxtop induction cooktop 9600LS

² Time ranges for water boiling have been opportunely verified for reasonable pot sizes and materials by means of the software Cook-STePS (Lombardi, Colombo, & Colombo, 2017).

Table 3

Summary of cooking habits in Mama Kevina with corresponding timings and induction cooker power ranges (LP: low power; MP: medium power; HP: high power).

	Timing [min]	Power range
Breakfast		
Water boiling	18–25	HP
Porridge	15–20	LP
Lunch/Dinner		
Ugali	20–30	LP
Rice	20–30	LP
Soup	20–40	LP
Vegetables/potatoes/meat/fish	5–10	MP
Water boiling	18–25	HP

soup cooking in the hot water for 20 to 40 min at LP. Accordingly, lunch or dinner total preparation time will vary between 40 and 75 min, encompassing a wide range of natural variability (Fig. 4).

Nevertheless, a local cooking habit is to prepare lunch right after breakfast in order to take it away for the working day – kept warm in specific insulated pots–, where most of the people work (Tanzania Food and Nutrition Centre and TAHEA Iringa, 2013). We reproduce this habit by combining a breakfast and a lunch/dinner cycle into a “full-preparation” cycle, considering a delay of 1 to 15 min between the two phases to consider the time required for the consumption of breakfast and for generic preparation tasks. Fig. 5 shows an example of this full-preparation cycle.

Micro-grid optimisation

Considering that the aim of the present study is to perform an accurate assessment of the techno-economic potential of e-cooking, we avoid the use of self-developed optimisation models that would hinder comparability with existing studies, rather relying on the state-of-the-art and widely adopted software for micro-grids optimisation HOMER Energy (HOMER Energy LLC, n.d.). Furthermore, the high detail of technology parameters that can be implemented into the software – as compared to the existing alternatives, such as iHOGA, DER-CAM, etc. (Dufo-López, Cristóbal-Monreal, & Yusta, 2016; Fathima & Palanisamy, 2015; Mashayekh, Stadler, Cardoso, & Heleno, 2017) – allows us to accurately simulate and compare actual commercial technologies based on manufacturers' datasheets.

The results of HOMER optimisation include the optimal size of each technological component and the resulting Levelised Cost of Electricity (LCOE) of the system. The latter is computed as in Eq. (1).

$$\begin{cases} LCOE = \frac{NPC}{E_{TOT}} \\ NPC = \sum_{y=1}^{LT} \frac{Inv(y) + O\&M(y)}{(1+r)^y} \\ E_{TOT} = \sum_{y=1}^{LT} \sum_{d=1}^{365} E_{d,y} \end{cases} \quad (1)$$

where:

$y = 1, \dots, LT$ represents each year of the plant lifetime LT ; $Inv(y)$ is the yearly investment and replacement costs of the components of the system; $O \& M(y)$ are the operation and maintenance yearly costs; $(1+r)^y$ is the discount factor, with r the real interest rate; $E_{d,y}$ is the daily energy calculated from the load profiles derived with the modified version of LoadProGen.

Load profiles generated by our novel script are imported into the software as a set of 365 daily load curves (each consisting of an independent randomisation). Solar resource data and ambient temperature data are obtained, respectively, from the NASA surface meteorology and solar energy database and from the MERRA-2 global dataset (freely available online at Renewables Ninja (Pfenninger & Staffell, 2016)).

The allowed system configuration comprises PV panels, batteries integrated with the solar charge controller and the inverter (Fig. 6). Batteries represent the most critical, sensitive and expensive system component; as such, we rely on HOMER's “Advanced Storage Module” (ASM) add-on in order to increase the detail in the characterisation of the electric behaviour and operation of the storage technology. For instance, the ASM allows taking into account series resistance, temperature effects on capacity, temperature effects on degradation rate, and cycle-by-cycle degradation based on depth of discharge.

Further details regarding the commercial components and the economic parameters used as a reference for the analysis are reported in Appendix A.

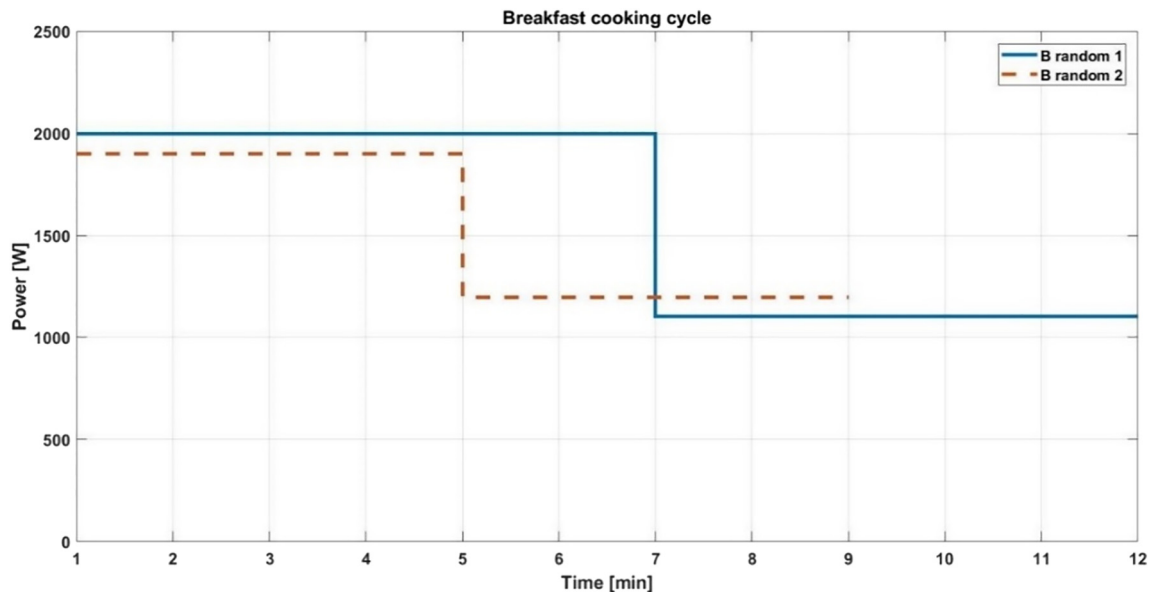


Fig. 3. Example of two different randomisations of the breakfast cooking cycle. Each of the two cooking tasks composing the cycle is subject to random variations of their duration and absorbed power, resulting also in a variation of the total cycle duration.

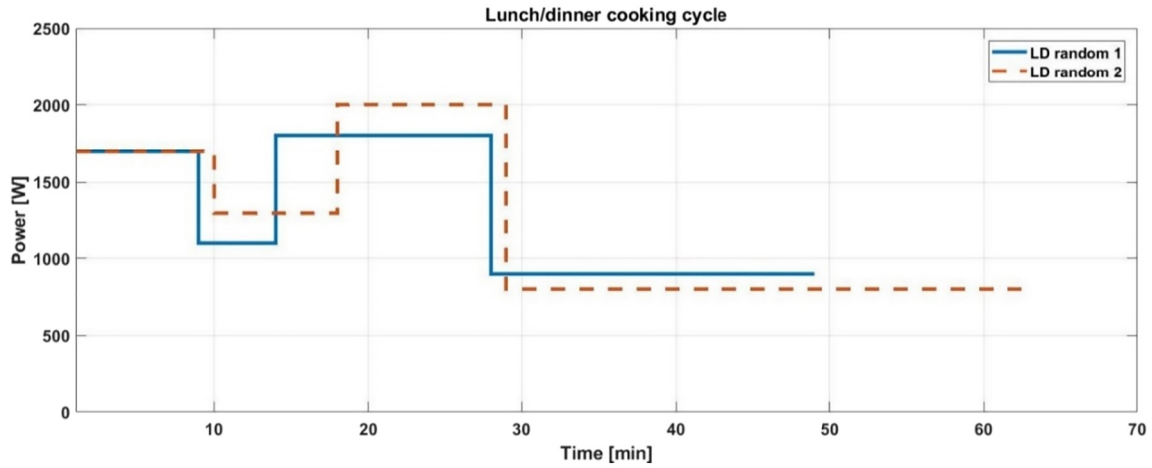


Fig. 4. Example of two different randomisations of a lunch/dinner cycle.

Scenarios definition

Table 4 reports the scenarios considered for the energy optimisation of the micro-grid. As a reference scenario, we assume that a micro-grid is installed to satisfy only conventional electric loads, such as lighting or basic entertainment (TV, radio), or those already covered by diesel generators; we call this scenario *Base load*. Conversely, we define a *Full cooking* scenario in which a complete transition from traditional biomass cooking to e-cooking takes place, i.e. induction cookers are installed within the micro-grid so as to replace 100% of existing cooking energy demand and are fully adopted by all users.

In addition to these two main scenarios, we also define two “off-design” scenarios that allow simulating possible divergences from an ideal user behaviour, namely: (i) a *Fuel stacking* scenario, and (ii) a *Behavioural change* scenario. The former assumes that only a certain share of households (0%, 25%, 50% and 75% shares are simulated) uses induction cookers for the preparation of all meals, while the remaining households use induction only for secondary meals preparation (e.g. milk, tea); this is done to check to what extent a fuel-stacking behaviour would impact the operation and cost of the e-cooking system. Such scenario is simulated only for the village case, considering that the centralised management of the Mama Kevina Hope Center should ensure that no fuel stacking takes place in the community-service case. The *Behavioural change* scenario, instead, assumes that the use of induction cookers makes unnecessary to prepare lunch immediately after breakfast – currently done to save time by avoiding double fire

preparation –, and it allows to shift the preparation of each meal just before each meal’s time. The latter scenario is simulated only for the community-service case, where the restricted number of people responsible for food preparation makes more likely such a demand-side management strategy. For these two off-design scenarios, the micro-grid size is constrained to remain the same as the optimal one obtained for the corresponding *Full cooking* scenario, while the operational strategy is allowed to be re-optimised by the software.

Techno-economic performance indicators

A comparison of the least-cost optimised e-cooking solution against existing technological alternatives (firewood, charcoal, kerosene, LPG) is performed by relying on the Levelised Cost for Cooking a Meal (LCCM), recently defined by Fuso Nerini et al. (Nerini et al., 2017) as the cost for cooking a “standard” meal with a given fuel-technology combination (Eq. 1).

$$LCCM = LCCM_{fuel} + LCCM_{stove} = \frac{C_{fuel} \cdot E_u}{\eta_{stove}} + \frac{\sum_{y=1}^n \frac{C_{stove}(y) + O\&M_{stove}(y)}{(1+r)^y}}{\sum_{y=1}^n \frac{MI(y)}{(1+r)^y}}$$

where:

- C_{fuel} is the fuel price in USD/kWh (for e-cooking, obtained as a result of the micro-grid optimisation);

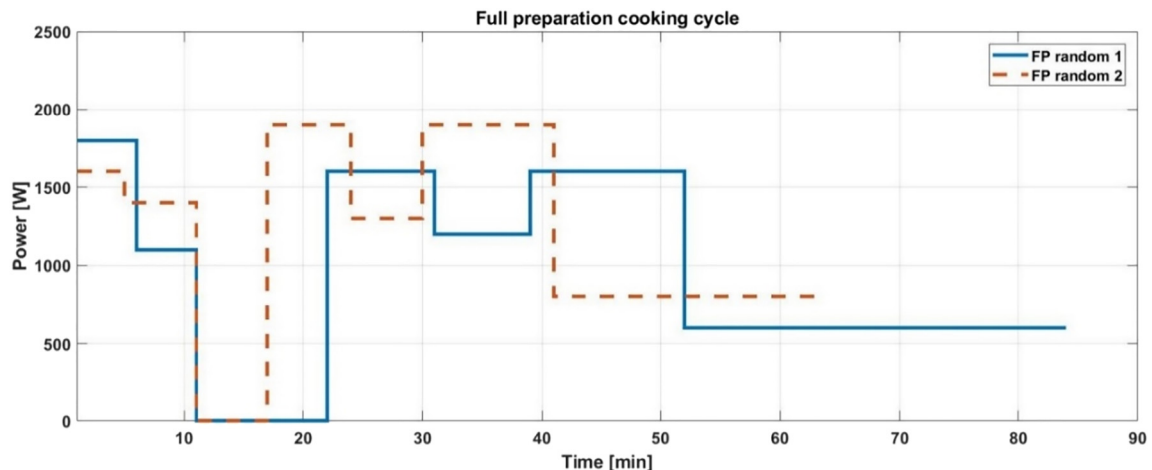


Fig. 5. Example of two different randomisations of a full-preparation cooking cycle. The delay between the two phases (breakfast and lunch) is also subject to randomisation.

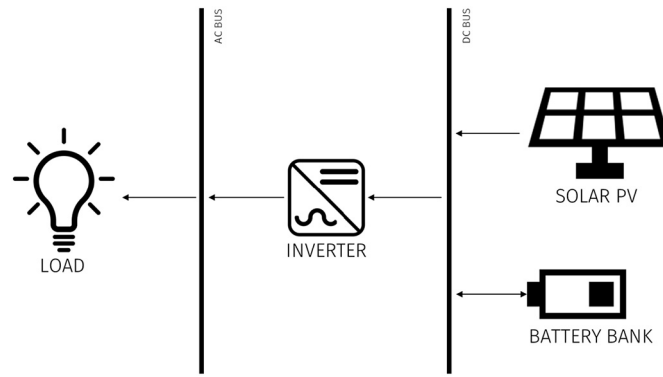


Fig. 6. Configuration scheme of the energy system.

- E_u is the average useful energy required for cooking a meal in kWh/meal;
- η_{stove} is the stove efficiency;
- $C_{stove}(y)$ is the stove purchase cost, occurred in the y -th year, in USD;
- $O \& M_{stove}(y)$ are the operation and maintenance yearly costs related to the stove;
- $Ml(y)$ is the amount of meals cooked in the y -th year;
- n is the stove lifetime in years.

The parameter E_u is assessed by considering that: firstly, the daily useful energy for cooking (i.e. energy that goes into the pot and the food) is the product of the modelled electric energy consumption for cooking and the efficiency of the induction stove; and secondly, the average number of “fully cooked” meals – as defined by Fuso Nerini et al. (Nerini et al., 2017) – for the two case studies is, respectively, 2.5 and 3, in accordance with how we defined typical meals in the [Cooking devices and cooking habits characterisation](#) section. Considering that we defined lower and upper bounds of stochastic variability for the computation of cooking events duration (hereafter named *fast cooking* and *slow cooking*), it is possible to define a range of corresponding extreme values for E_u .

Other parameters that are subject to variability depending on context-specific variables and natural fluctuations are C_{stove} and η_{stove} (except for e-cooking, for which C_{stove} is an endogenous optimisation parameter and η_{stove} is set at 90%). Combining all the sources of variability it is possible to generate two extreme LCCM values for each fuel-technology combination, as the product of: a) fast cooking, higher efficiency range, and lower fuel price range; or b) slow cooking, lower efficiency range, and higher fuel price range.

Table 4
Scenarios definition.

Scenarios	Load to be supplied by the electric power system
Based load	– Conventional electric load only
Full cooking	– Conventional electric load – e-cooking load: <i>complete</i> transition from traditional biomass cooking to e-cooking by keeping the same cooking habits
Fuel stacking	– Conventional electric load – e-cooking load: <i>gradual</i> transition from traditional biomass cooking to e-cooking by keeping the same cooking habits: only a certain share of households (0%, 25%, 50% and 75%) uses induction cookers for the preparation of all meals, while the remaining households use induction only for secondary meals preparation (e.g. milk, tea)
Behavioural change (only for the community-service)	– Conventional electric load – e-cooking load: <i>complete</i> transition from traditional biomass cooking to e-cooking by changing the cooking behaviours along the day.

We highlight that the nominal discount rate is set at 9% (Leach & Oduro, 2015), while the inflation rate is set at 5% (Zubi, Spertino, Carvalho, Adhikari, & Khatib, 2017). For each component, replacement and O&M costs are considered for a project lifetime of 20 years. Replacement costs have been evaluated equal to the purchase costs in order to not make assumption about future price development. Reference values for O&M costs are taken from the Fuso Nerini et al. (2017) and World Bank reports (Tenenbaum, Greacen, Siyambalapitiya, & Knuckles, 2014). For the PV modules, MPPT solar charge controllers and system inverter O&M costs are considered equal to 1.5% of the specific purchase cost. For batteries, instead, 22 USD/kWh per year is taken as a reference value for lead-acid technology. In addition, we consider BOS costs, such as mounting, installation, cabling and infrastructure. In particular, we assumed costs of 90 USD/kW for mounting, 60 USD/kW for installation, 60 USD/kW for wirings and 55 USD/kW for infrastructure. It is worth specifying that a 5% annual capacity shortage is considered acceptable, so as to avoid a sub-optimal sizing based on the worst conditions of the year in terms of peak load or unfavourable external conditions.

Results and discussion

Simulated loads

Residential case study

The results obtained by simulating 365 curves with the stochastic load model are reported in Fig. 7, for the two main scenarios (*Base load* and *Full cooking*). It is worth noting that the maximum peak power of the load curve passes from around 9 kW to around 44 kW for the *Full cooking* scenario, confirming that the introduction of induction cookers within micro-grids conceived for powering only simple appliances represents a radical change in the power that the system needs to supply.

Fig. 8, instead, shows how peaks may be reduced as a consequence of a *Fuel stacking* behaviour in the village case, from complete fuel stacking (i.e. 100% of household use induction only for minor meals) to increasing penetration of complete adoption of e-cooking. These curves are used to check how much an underuse of e-cooking due to fuel stacking may impact the techno-economic performance of a residential-scale micro-grid that is optimised for *Full cooking*.

Community-service case study

As shown in Fig. 9, in the community-service case, the introduction of e-cooking generates a reduced increase in the peak load given the pre-existing presence of energy-intensive appliances (such as a washing machine) and to a sort of economy of scale ensured by the centralised cooking process.

The off-design *Behavioural change* scenario hypothesised for the community service is instead represented by the load in Fig. 10, where a new peak appears at lunch time while the morning peak is reduced. Looking at the average profile, it is worth noting that the maximum

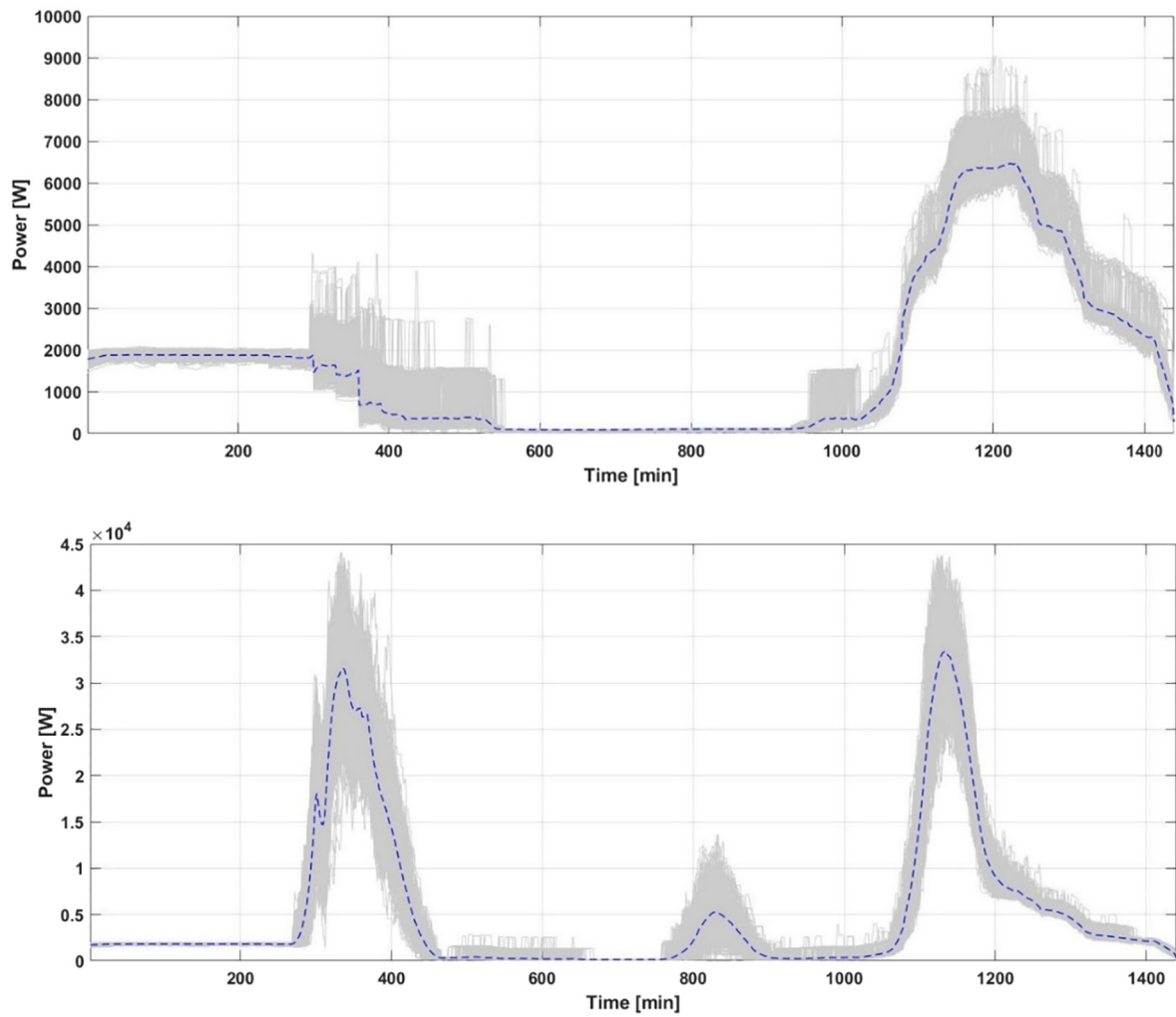


Fig. 7. Base load (top side) and Full cooking (bottom side) load curves of the residential case study.

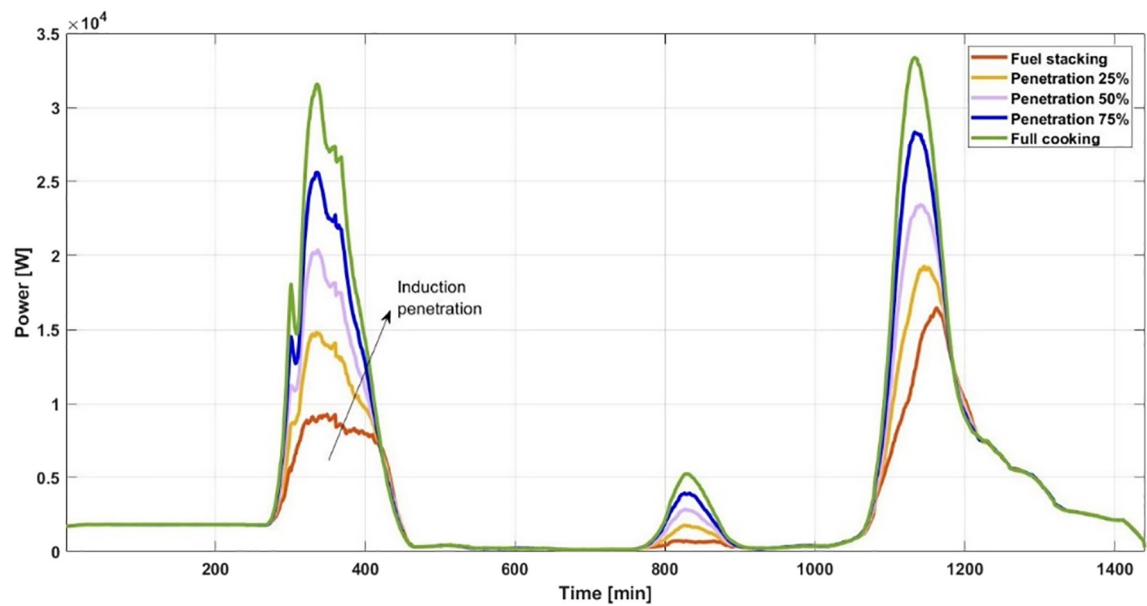


Fig. 8. Average curves of the Fuel stacking scenarios.

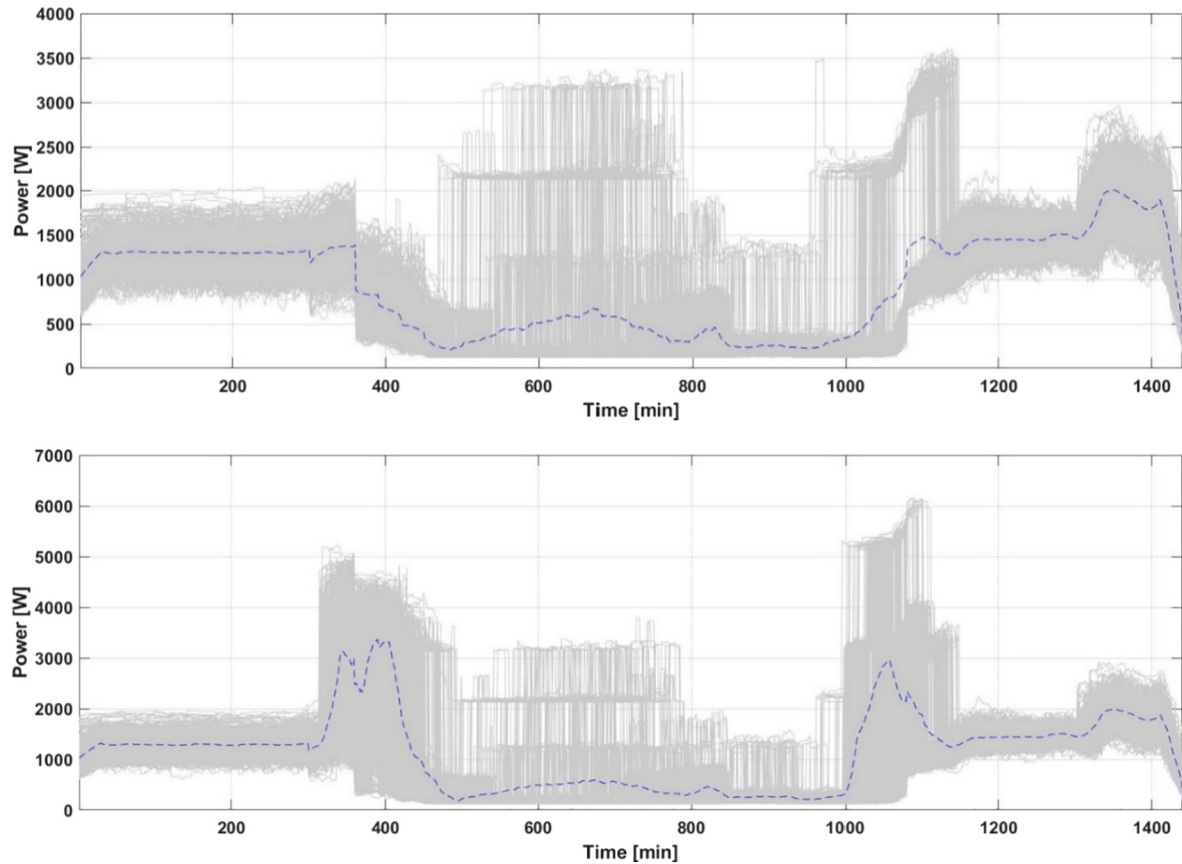


Fig. 9. Base load (top side) and Full cooking (bottom side) load curves of the community-scale case study.

daily peak load is only slightly reduced by this potential demand-side management intervention.

Optimal system configuration and off-design response

Residential case study

The least-cost optimised micro-grid size, for the residential case, is reported in Table 5 for the Base load and the Full cooking scenarios. As expected from the corresponding huge increase in the peak load, a Full cooking scenario requires a significant increase in the size of the system; the PV and battery capacity more than triples, along with the inverter size. This implies, of course, a significant rise of the NPC, while the LCOE increases only slightly due to the parallel increase

of the load demand. In short, even though the cost of electricity remains similar, the required capital investments are significantly higher.

When assuming a Fuel stacking scenario, instead, the average daily energy demand decreases from 123.3 kWh to 97.5 kWh per day (assuming, as a reference example, a 50% fuel stacking), with implications on batteries operation and on the computation of the LCOE. In fact, the autonomy of the battery bank increases from 70.1 to 88.7 h, and the expected battery lifetime grows from 15.4 years to 18.8 (considering 57 cycles per year and 80% maximum DOD accepted). As the micro-grid components are the same as the Full cooking scenario, capital costs remain unchanged, but the increased lifetime of the battery pack allows postponing its own replacement, thus resulting in a higher

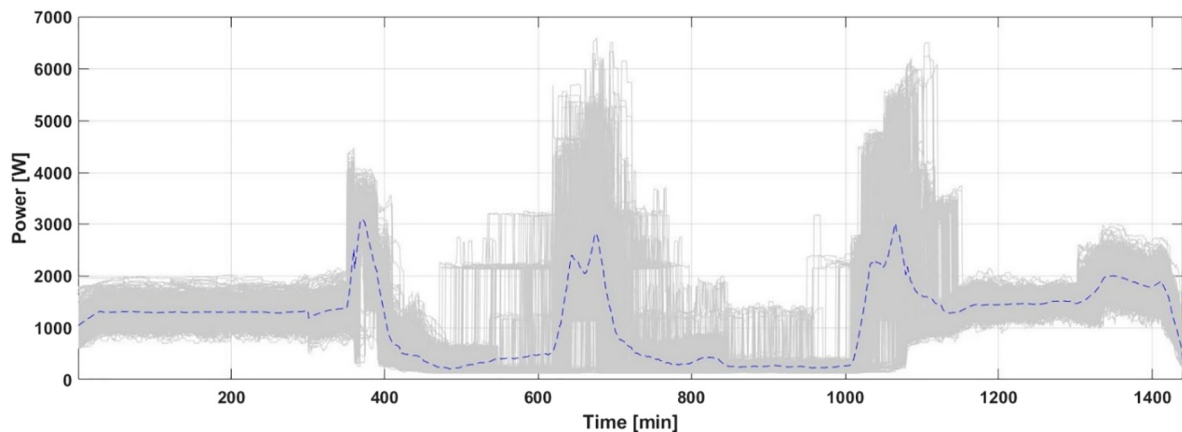


Fig. 10. Load curves of the Behavioural change scenario for the community-scale case study.

Table 5

Results of the micro-grid optimisation for the residential scale case study.

	Base load scenario	Full cooking scenario
PV capacity [kW]	12	37
Battery capacity [kWh]	104	417
LCOE [USD/kWh]	0.23	0.24
NPC [USD]	43 771	136 547

salvage value. For this reason, the NPC reduces to 127'525 USD, while the LCOE increases (up to 0.259 USD/kWh) as a consequence of the reduced energy production that overwhelms the cost savings. Nonetheless, the cost remains much higher than for the *Based load* scenario; this is due to the fact that the micro-grids is optimised only for basic residential loads, which are too low compared with the nominal power required by induction cookers.

Community-service case study

For the community-service case, the additional PV and battery capacity needed to supply the cooking loads is not so relevant (Table 6), thus resulting in just a minor increase in the NPC of the system, mainly associated with the capital cost of the small additional capacity installed. As a result, the LCOE is reduced, driven by an increase in the energy generation that overcomes the small cost increase.

Furthermore, the simulation of the potential *Behavioural change* scenario results only in a very slight reduction in the number of batteries equivalent cycles per year (from 65 to 66), not affecting the estimated battery lifetime of 12 years. In fact, the daily energy demand absorbed from the storage remains almost unchanged, even though peaks are shifted in terms of time frames, and temperature plays in this case the driving battery degradation factor.

Levelised Cost for Cooking a Meal (LCCM)

Fig. 11 shows the range of LCCMs obtained for the various cooking technologies considered. The e-cooking option proves to be quite competitive, with an upper range limit similar to charcoal and lower than kerosene, while the lower limit lies within the ranges of all other cooking options, including firewood. Interestingly, LPG seems to be the most expensive alternative and less competitive than e-cooking.

Fig. 11 shows also the corresponding LCCM values calculated, for each technology, by Fuso Nerini et al. (2017) for a similar case: the latter lie in all cases within the range calculated in the present study; however, the e-cooking value is particularly close to upper bound resulting from our analysis. This underestimation of the e-cooking potential, in Nerini et al., is arguably attributable to site-specific conditions and to different assumptions regarding the type of electric stove adopted, but also to the lack of a detailed and cost-optimised analysis of the micro-grid system supplying the electricity. As such, this difference highlights the importance of taking into account highly-performing and cost-optimised electricity supply configurations when considering e-cooking, as proposed by the present study.

Similar results are obtained for the community-service case (Fig. 12) with some minor differences due to the different conditions and parameters of this second case-study, which is also not comparable to any other previous study in the literature.

Table 6

Results of the micro-grid optimisation for the community-scale case study.

	Base load scenario	Full cooking scenario
PV capacity [kW]	8	9
Battery capacity [kWh]	69	85
LCOE [USD/kWh]	0.27	0.26
NPC [USD]	29'441	34'412

Conclusion

According to the 7th Sustainable Development Goal of the Agenda 2030, the current 2.8 billion people who lack access to clean cooking facilities should ideally achieve it by 2030. Given the environmental and health drawbacks entailed by improved biomass stoves as well as the climate concerns over the potential widespread adoption of LPG-fuelled stoves, electric cooking (i.e. e-cooking) based on renewable sources and highly-efficient cooking devices could represent a sustainable and reliable alternative to meet this goal. Our study investigated the e-cooking feasibility through: (1) the development of an appropriate model for the assessment of electric cooking load profiles for a residential and a community-service case study in Tanzania; and (2) a subsequent least-cost optimisation of the related power supply system, i.e. a PV-battery micro-grid.

The results show how, for the residential case (i.e. within a system characterised by only basic electricity loads), the introduction of induction cookers produces remarkable changes of the overall electricity load profile, giving rise to two steep demand peaks in the early morning and in the evening. On the contrary, the inclusion of the same technology within a community service already characterised by the presence of highly energy-intensive appliances does not produce as much significant changes in the peak demand.

Such results impact, in turns, on the cost-optimal size of the micro-grid for the two cases: for the residential case, the introduction of induction cookers requires a power capacity and a related investment which are triple compared to the baseline case with no e-cooking (even when considering potential fuel stacking practices). Conversely, for the community-service case, the adoption of induction cookers entails only a 17% higher installed capacity and capital investment. Furthermore, both case studies demonstrate how the inclusion of electric cooking in the analysis does not produces appreciable changes in the LCOE compared to a system without electric cooking. These results point out an important consideration: since the mechanism of tariff creation is usually based on the LCOE, the introduction of e-cooking does not significantly change the cost of electricity, which is an important parameter for the long-term sustainability of access to electricity programmes.

The LCCM analysis confirms that highly-efficient e-cooking can be absolutely cost-competitive with other modern-fuel alternatives (such as LPG), while the comparison against results obtained by a similar study show that e-cooking profitability can be much higher when considering an electricity supply ensured by a cost-optimised modern energy system configuration (e.g. hybrid micro-grid) rather than a simple standalone PV-battery system.

Future works shall consider the investigation of the e-cooking concept on a village-scale system characterised by both residential and productive uses of energy, as well as combining the generation of load profiles with behavioural models aimed at assessing the feasible penetration of e-cooking in rural communities which are usually characterised by high fuel stacking behaviours.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esd.2019.01.005>.

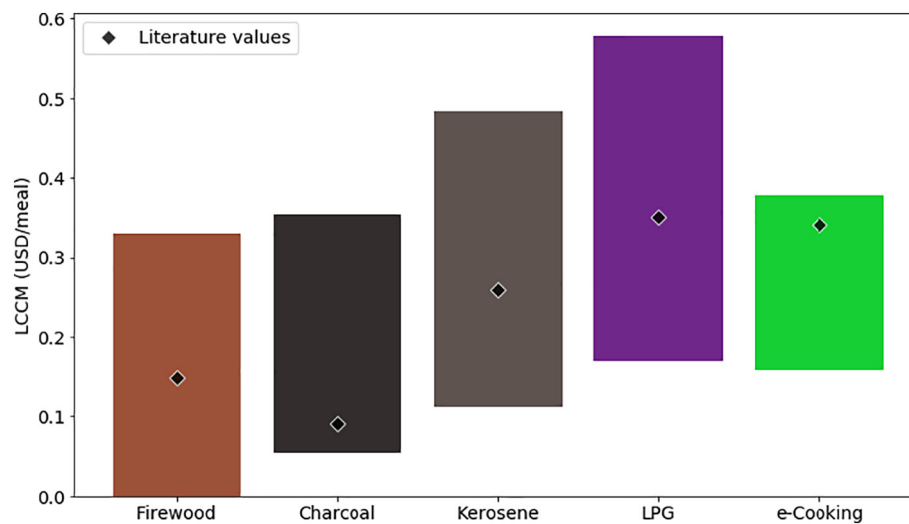


Fig. 11. Ranges of LCCMs for the various cooking technologies considered in access to energy programmes, for the residential case study. Black markers represent the values calculated by Fuso Nerini et al. (2017) for a similar case, used for comparison.

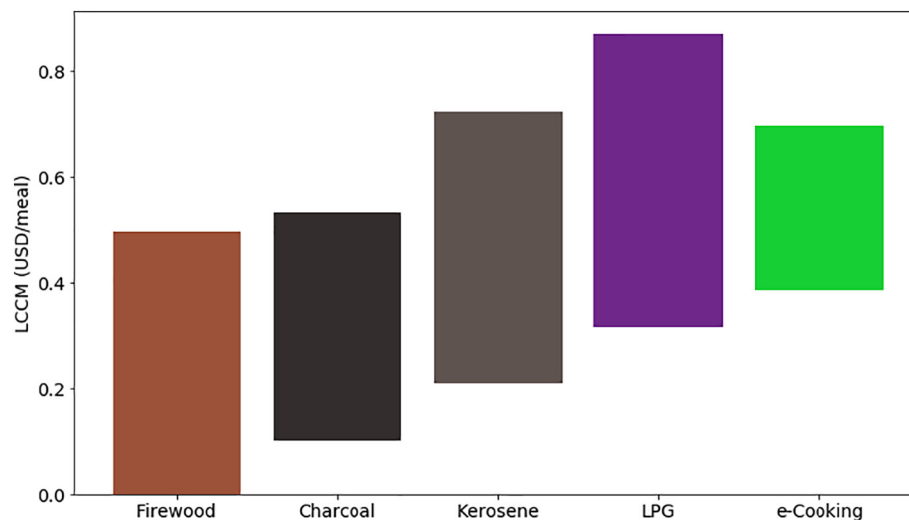


Fig. 12. Ranges of LCCMs for the various cooking technologies considered in access to energy programmes, for the community-service case study.

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